Optimizing Metro Dispatch Scheduling for Profit Maximization: A Case Study using Bangalore Namma Metro

A Linear Programming Approach to Urban Transit

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1 Problem

Urban mass transit systems, while critical for city infrastructure, frequently operate at a severe financial deficit. They are historically designed around passenger convenience—running high-frequency schedules that often result in half-empty trains during off-peak hours, incurring massive operational costs.

Research Problem (RP): The Bangalore Metro Rail Corporation Limited (BMRCL) operates a fleet of 58 trains across its Purple and Green lines. The current operational model relies on static timetables with rigid, high-frequency dispatches (approximately 800 total daily trips). Given the high operational cost per trip, there is a fundamental tension between maintaining civic convenience (short wait times) and achieving financial sustainability (running at optimal capacity).

Research Question (RQ): What is the mathematically optimal, time-varying train dispatch schedule across the Namma Metro network that maximizes expected daily operational profit, without violating physical signaling constraints, rolling stock limits, or minimum public service mandates?

2 Plan

To answer the research question, we modeled the transit system as a convex optimization problem. Because the revenue generated per passenger and the cost incurred per train scale linearly, Linear Programming (LP) is the ideal mathematical framework.

Measurement and Methodology: The objective is to maximize the expected daily profit. Profit is derived from the ticket revenue of passengers served minus the operational cost of the dispatched trains. To ground the model in reality, the core constraint is the passenger demand.

Data Collection and Sampling: We utilized secondary, empirical footfall data provided by BMRCL for the entire month of August 2025. The dataset acts as a census of all smart-card and token taps across all 69 operational stations. By aggregating this historical data, we can calculate the expected value (mean) of the random variable representing passenger footfall for any specific hour on a typical business day, providing the necessary probabilistic parameters for our optimization constraints.

3 Data

The raw dataset consisted of daily station-wise entry/exit records for August 2025. Preparing this data for the optimization algorithm required several aggregation steps:

1. **Line Mapping:** The 69 individual stations were grouped into two vectors representing the operational lines: the Purple Line (P, 37 stations) and the Green Line (G, 32 stations).

2. **Temporal Isolation:** Aggregate columns (e.g., cumulative daily totals) were removed. The data was filtered to isolate the 24 specific operating hour blocks (from 00:00 Hrs to Midnight), ensuring a complete daily cycle was captured including late-night stragglers and early-morning dead zones.

3. **Expectation Calculation:** For every operating hour t and line l, the total footfall was summed across the month and divided by the number of unique business days. This yielded $D_{l,t}$, a continuous vector representing the Expected Hourly Passenger Demand. No significant outliers were removed, as massive demand spikes (e.g., due to sporting events) are valid reflections of real-world variance.

4 Analysis

4.1 Descriptive Statistics

Before optimizing the dispatch schedule, a statistical analysis of the expected demand vectors ($D_{l,t}$) was conducted to understand system variance. Table 1 summarizes the core parameters of the expected passenger footfall for both the Purple and Green lines across a complete 24-hour operational cycle.

Table 1: Summary Statistics for Expected Hourly Demand

Line	**Total Daily**	**Mean Hourly**	**Peak (Max)**	**Off-Peak (Min)**	**Std. Dev.**
Purple	321022.4	13375.93	32717.28	0	10596.12
Green	244436.6	10184.86	20765.22	0	7507.32

The data reveals severe temporal and spatial asymmetry. As illustrated in Figure 1, the system's demand is heavily bimodal. Following the system dead-zone (01:00 to 04:00 Hrs), footfall spikes aggressively during the morning (08:00–10:00 Hrs) and evening (17:00–20:00 Hrs) commutes.

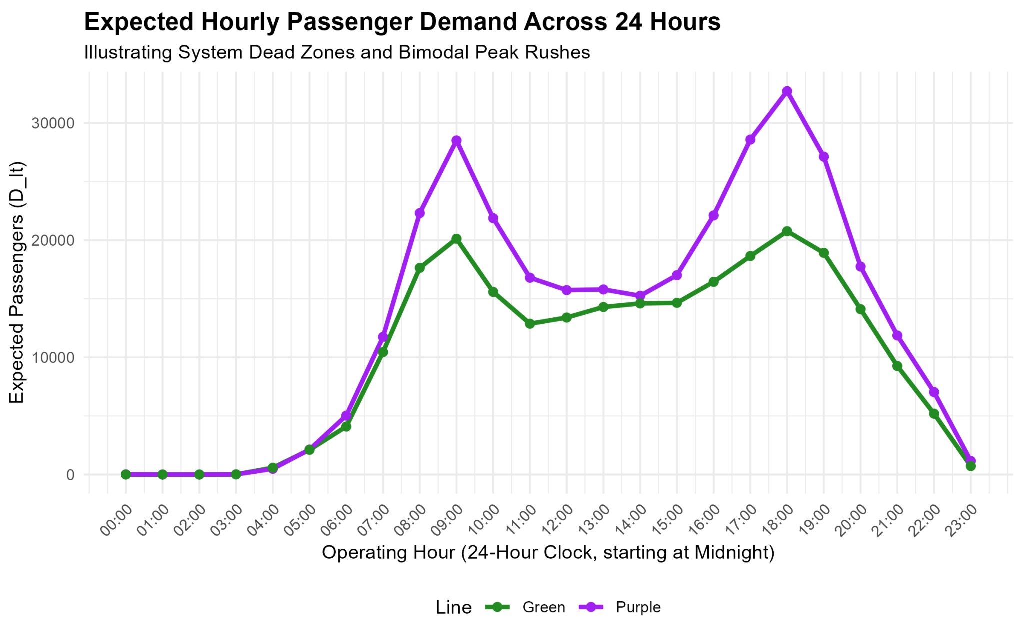

Figure 1: Time-Series of Expected Passenger Demand demonstrating system dead zones and extreme bimodal peak rushes across 24 hours.

Furthermore, the boxplot in Figure 2 demonstrates the spatial variance between the two lines. The Purple Line exhibits a significantly higher median demand and a much wider Interquartile Range (IQR) than the Green Line. The extreme top whisker of the Purple Line highlights maximum peak outliers exceeding 30,000 passengers per hour. This massive variance mathematically invalidates the financial efficiency of a static, flat-rate dispatch schedule, directly necessitating a dynamic optimization approach.

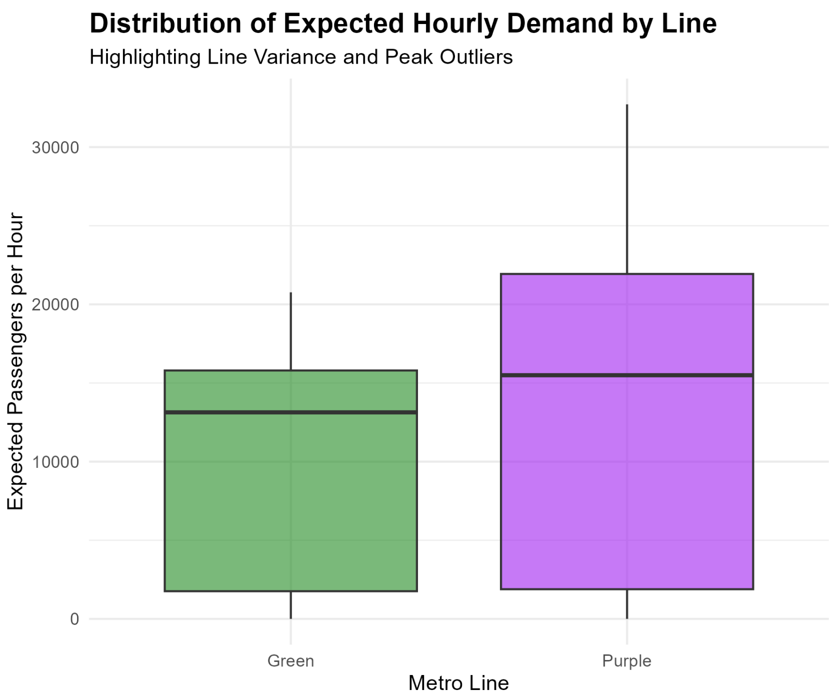

Figure 2: Boxplot detailing the spread, median disparity, and peak outliers of hourly demand between the Purple and Green lines.

4.2 Mathematical Formulation

The problem was formulated as a Linear Program to maintain convexity. Let $l \in \{P, G\}$ denote the metro lines, and let $t \in \{0, 1, \ldots, 23\}$ denote the 24 specific operational hours of the day (starting at midnight).

Decision Variables (Continuous):

- $x_{l,t}$: Frequency of trains dispatched on line l during hour t.
- $y_{l,t}$: Number of passengers successfully served on line l during hour t.

Parameters (Constants):

- $R = 27$: Average revenue per commuter (₹).
- $C = 22818$: Operational cost per train trip (₹).
- $K = 2068$: Maximum physical crush capacity of a 6-coach train.
- $F = 58$: Maximum rolling stock (total fleet available).
- $D_{l,t}$: Expected random passenger demand on line l during hour t.

Objective Function (Expectation Type): We aim to maximize the expected daily profit $E[Z]$:

$$\text{Maximize } E[Z] = \sum_{t=0}^{23} \sum_{l \in \{P,G\}} (27 \cdot y_{l,t} - 22818 \cdot x_{l,t}) \tag{1}$$

Constraints:

$$y_{l,t} \leq 2068 \cdot x_{l,t} \quad \forall l,t \qquad \text{(Physical Capacity)} \quad (2)$$
$$y_{l,t} \leq D_{l,t} \quad \forall l,t \qquad \text{(Expected Demand Limit)} \quad (3)$$
$$x_{P,t} + x_{G,t} \leq 58 \quad \forall t \qquad \text{(Fleet/Rolling Stock Limit)} \quad (4)$$
$$x_{l,t} \leq 20 \quad \forall l,t \qquad \text{(Signaling Safety Upper Bound - 3 min headway)} \quad (5)$$
$$x_{l,t} \geq 3 \quad \forall l,t \qquad \text{(Minimum Civic Mandate - 20 min headway)} \quad (6)$$
$$y_{l,t} \geq 0 \quad \forall l,t \qquad \text{(Non-negativity)} \quad (7)$$

4.3 Optimization Methodology

By keeping the decision variables (x, y) continuous, the constraints form a bounded convex polyhedron. The problem was computationally solved via the Simplex algorithm utilizing the `lpSolve` package in R.

4.4 Discussion of Findings

The solver identified a global optimum yielding an Expected Daily Profit of **₹8,102,022.41**. The optimal schedule generated **₹15,267,393.00** in revenue against **₹7,165,370.59** in operational costs.

To achieve this, the solver allocated a total of $\sum x_{P,t}$ = **174.97** daily trips to the Purple line and $\sum x_{G,t}$ = **139.06** daily trips to the Green line. The algorithm successfully shifted the 58-train fleet dynamically. It allocated heavy resources to the Purple line during the asymmetrical peak hours highlighted in Figure 1 to capture maximum revenue, while dropping strictly to the minimum safety mandate (3 trains/hr) during the 01:00 to 04:00 Hrs dead zones to preserve operational capital.

5 Conclusion

The optimization model successfully found the mathematical global maximum for transit profitability, but in doing so, revealed a profound discrepancy between mathematical optimization and civic utility.

Interpretations & New Ideas: The optimal LP schedule recommended mathematically optimizing the trips heavily downward compared to BMRCL's current real-world frequency (which is approximately 800 daily trips). The LP solver, bound only by a monetary objective function, ruthlessly eliminated "convenience trips," forcing passengers to wait longer so that every dispatched train operated at its absolute maximum capacity (2068 passengers). This demonstrates that BMRCL operates with an unquantified "hidden constraint"—civic convenience—which causes them to operate at a massive financial deficit compared to the mathematical ideal.

Limitations: The primary limitation of this study is the purely financial nature of the objective function. Because "customer satisfaction" and "wait time" were not quantified as negative cost coefficients, the algorithm actively sacrificed commuter comfort for operational efficiency.

Future Directions: In future iterations, this model should be advanced from a Linear Program to a Non-Linear Program. By introducing an exponential "Wait-Time Penalty" coefficient into the objective function (where profit is penalized for every minute a passenger waits over 5 minutes), the model could find the true Pareto-optimal balance between corporate financial sustainability and public passenger satisfaction.

Individual Contributions

- **Gagan Krishna S (BSDBG2506):** Led the mathematical formulation of the Linear Programming constraints, ensuring strict adherence to signaling physics and fleet limitations, and drafted the final interpretations.
- **Garg Parashar (BSDBG2507):** Managed the 'Data' phase, executing the R scripts required to clean the raw BMRCL dataset and calculate the expected demand vectors ($D_{l,t}$).
- **Saurav Kumar (BSDBG2517):** Handled the computational 'Analysis' phase, implementing the `lpSolve` Simplex algorithm to navigate the convex polyhedron and extract the optimal values.